

Idea: StayOps — Relocation Accommodation Operations Platform

Problem

Small relocation and accommodation agencies manage enquiries, room availability, temporary holds, landlord approvals, resident documents, move-in preparation, maintenance, rent records and landlord payments across disconnected spreadsheets, email threads and messaging applications. This fragmentation can cause double bookings, expired holds remaining active, missed preparation tasks, unclear payment records and poor visibility across agency-owned and partner-supplied properties.

The problem is broader than ordinary property management. A relocation agency must match a person to suitable temporary or medium-term accommodation, coordinate approval from an external landlord, prepare the room before arrival and manage changes such as delayed arrivals, extensions and transfers.

Evidence

A commercial software category already exists for relocation management, temporary housing and corporate accommodation, demonstrating that organisations purchase technology to source accommodation and manage bookings and stays. Dedicated co-living platforms also manage room- or bed-level inventory, individual contracts, occupancy and maintenance, validating the operational need at property level.^{[1][2][3][4][5][6]}

Existing property-operations products emphasise that room readiness depends on coordinated housekeeping, inspections, maintenance, access and vendor work rather than a booking calendar alone.^{[7][8]}

Proposed Product

StayOps is a multi-tenant SaaS platform for local relocation and accommodation agencies managing their own rooms alongside properties supplied by external landlords. It supports the full operational lifecycle:

Accommodation request → matching → temporary hold → landlord approval → booking → documents and deposit → room preparation → move-in → resident support → extension, transfer or move-out.

The product contains six integrated components:

- Agency, landlord and property administration.
- Transaction-safe availability and booking management.
- Constraint-based accommodation matching.

- Resident and contractor applications.
- Room-readiness and maintenance workflows.
- Payment recording and landlord reconciliation.

Target Customers

- Small and medium-sized relocation agencies.
- Corporate accommodation providers.
- International-student accommodation agencies.
- Co-living operators.
- Employers arranging medium-term accommodation for relocated staff.
- Property managers operating room-by-room rentals.

Users

User	Main Activities
Agency administrator	Configures organisations, users, landlords, properties and operational rules
Placement officer	Records accommodation requests, reviews matches, places holds and confirms bookings
Property coordinator	Manages inspections, cleaning, maintenance and room readiness
Landlord	Reviews placement requests, approves bookings and views statements
Resident	Completes onboarding, uploads documents, views booking details and reports issues
Contractor	Accepts tasks, records visits and submits completion evidence
Finance officer	Records payments, reconciles deposits and prepares landlord settlements

Existing Competitors

Existing alternatives include global relocation and corporate-housing platforms such as ReloQuest and AltoVita, local or specialised accommodation systems such as Studentpad, and co-living platforms such as HabitaMax, ColivHQ and Booking Ninjas.^{[9][2][3][4][6][1]}

The project must not claim that accommodation or property-management software is new. Its contribution is a tightly defined workflow for small local agencies operating mixed accommodation inventory.

Specific Competitive Gap

A relocation-accommodation operating platform for small local agencies managing both agency-controlled and partner-landlord rooms, combining explainable placement matching, temporary availability holds, operational room readiness and landlord reconciliation.

Many property systems begin after a tenant has already selected a property. StayOps begins with the accommodation request and manages the uncertain period before confirmation, including ranked offers, expiring holds, landlord approval and alternative placements. Unlike a

basic booking platform, a room is not marked ready until its required inspection, cleaning, repair, inventory and access tasks are complete.

Six-Person Technical Architecture

Team Member	Major Subsystem	What They Actually Code	Technical Difficulty
1	Core SaaS and Identity Platform	Multi-tenancy, authentication, role-based access control, agency and landlord accounts, property hierarchy, API gateway, tenant isolation and audit logging	High
2	Availability and Booking Engine	Room calendars, temporary holds, hold expiry, database locking, double-booking prevention, booking state machine, cancellations, extensions and room transfers	Very High
3	Placement and Matching Engine	Hard-constraint filtering, weighted preference scoring, ranked explainable matches, alternative recommendations and matching evaluation	Very High
4	Resident and Field Application	Responsive resident portal, document upload, onboarding, agreement workflow, move-in checklist, maintenance reporting, contractor interface and offline synchronisation	High
5	Operations Workflow Engine	Room-readiness workflows, cleaning and inspection tasks, contractor assignment, deadlines, escalation rules, notifications and event processing	High
6	Finance and Control Dashboard	Payment ledger, deposits, rent schedules, agency fees, landlord settlements, reconciliation workflow, occupancy analytics and real-time management dashboard	High

Every member owns a production subsystem with its own database logic, APIs, tests and user-facing integration. CI/CD, integration testing, security review, documentation and deployment are shared team responsibilities rather than one member's complete role.

Core State Machines

Booking Lifecycle

Enquiry → Matched → Offered → Held → Awaiting Approval → Confirmed → Occupied → Completed

Alternative transitions include:

- Offer rejected.
- Hold expired.
- Landlord rejected.
- Booking cancelled.
- Stay extended.
- Resident transferred.
- Resident departed early.

Room-Readiness Lifecycle

Occupied → Departure Pending → Inspection Required → Work Required/Cleaning Required → Verification Required → Ready

The booking engine may assign a future stay to a room, but move-in cannot be completed unless the readiness workflow reaches the **Ready** state.

Data Sources

- Property, landlord and resident data entered directly by a pilot agency with appropriate permission.
- Synthetic records for automated testing and load testing.
- OpenStreetMap for location display and approximate distance calculations.
- Imported CSV files for migrating existing property, room and booking records.
- Email or calendar integrations in the enterprise phase.

A real pilot should use only the minimum resident information necessary for the workflow. Test and demonstration environments should use fictional personal records.

Legal and Data-Access Risk

Risk level: Medium. The system may process resident names, contact details, identity documents, contracts and financial records. These are personal data and require a clear lawful basis, access controls, retention rules, encryption, audit logs, backups and procedures for data-subject requests.

The university MVP can reduce risk by avoiding passport storage, card processing and automated eligibility decisions. It should store only fictional resident information during development; a controlled pilot can use limited real data after the operator confirms its responsibilities and provides appropriate privacy information.

External Dependency Risk

Risk level: Low to Medium. A working MVP does not require government APIs, proprietary marketplace data or specialised hardware. It can operate using manually entered inventory, CSV imports and standard email notifications.

A pilot agency is strongly recommended because it turns the system into a validated product, but the architecture and demonstration can still be completed if the pilot withdraws. Payment gateways, digital-signature providers, accounting systems and external property portals should be optional integrations rather than core dependencies.

Professional-Expertise Requirement

No licensed professional is required to develop the platform. Agency staff and landlords remain responsible for accommodation approval, contractual terms, legal compliance, property standards and financial decisions. StayOps coordinates authorised workflows and records decisions; it does not provide legal advice or determine whether a person is eligible for housing.

Ethical Risks

- Exposure of resident identity or financial information.
- Unfair matching based on protected or sensitive characteristics.
- Excessive collection of documents.
- Staff viewing records outside their responsibilities.
- Incorrect availability causing a resident to expect accommodation that is not secured.
- Automated rejection without human review.

Mitigations include data minimisation, tenant-level isolation, granular permissions, encryption, access logs, human approval of placements, explainable matching factors and exclusion of protected characteristics from the matching score.

AI Role

AI is optional and must not be the main product. Possible non-critical uses include extracting structured fields from an accommodation request, categorising maintenance descriptions or drafting routine messages.

The accommodation matching engine should initially use deterministic constraints and transparent weighted scoring. This makes the result testable and allows staff to see why a property was recommended. Generative AI should never confirm or reject a placement autonomously.

Why AI Cannot Replace It

The product's core value comes from persistent inventory, transactional room holds, concurrent booking protection, workflow state transitions, permissions, document control, payment ledgers and operational accountability. A chatbot cannot reliably prevent two users from reserving the same physical room, release an expired hold atomically, reconcile a deposit or prove who approved a booking.

MVP

The 13-week MVP should include:

- One deployable multi-tenant web platform.
- Agency, placement-officer, landlord, resident and contractor roles.
- Buildings, units, rooms and bed-level inventory.

- Accommodation-request form.
- Availability search and explainable ranked matching.
- Temporary holds with automatic expiration.
- Double-booking prevention using database transactions.
- Landlord approval or rejection workflow.
- Booking confirmation, extension and room transfer.
- Resident onboarding and limited document upload.
- Move-in, inspection, cleaning and move-out checklists.
- Maintenance reporting and contractor assignment.
- Payment, deposit and fee recording without processing cards.
- Basic landlord settlement statement.
- Notifications and audit history.
- Occupancy and operations dashboard.
- Docker deployment, automated tests, monitoring and backup procedure.

Out of Scope

- Direct card or bank payment processing.
- Automated tenant screening.
- Legal-document generation presented as legal advice.
- Integration with government identity systems.
- Scraping accommodation marketplaces.
- Full accounting functionality.
- Native iOS and Android applications; a responsive PWA is sufficient.
- AI-based acceptance or rejection decisions.
- Large global accommodation marketplace functionality.

Enterprise-Scale Version

- Multiple agencies and geographical regions.
- White-labelled resident and landlord portals.
- Integration with accounting and CRM systems.
- Payment-gateway and open-banking reconciliation.
- Qualified digital signatures.
- Calendar, email and messaging integrations.
- Automated landlord statements and invoicing.
- Advanced contractor performance and service-level monitoring.

- Dynamic pricing and occupancy forecasting.
- Public API and partner webhooks.
- Data warehouse and portfolio analytics.
- Configurable workflows for corporate, student and temporary accommodation.

Business Model

- Monthly SaaS subscription based on active rooms or beds.
- Tiered plans for small agencies, growing operators and enterprise customers.
- One-time onboarding and data-migration fee.
- Optional charges for accounting, payment and messaging integrations.
- White-label and custom-workflow fees for larger operators.

The first pilot can be free for a limited period in exchange for workflow access, structured feedback and permission to use anonymised operational metrics in the academic evaluation.

13-Week Feasibility

Feasibility: High if the scope remains controlled. The team should build a modular monolith plus one separate matching or workflow service rather than six independently deployed microservices. This preserves individual ownership without creating unnecessary DevOps complexity.

Weeks	Main Deliverables
1-2	Pilot interviews, workflow mapping, requirements, architecture, security model and UI prototype
3-4	Identity, multi-tenancy, property inventory, database schema and API contracts
5-6	Availability, temporary holds, booking state machine and initial matching engine
7-8	Landlord approval, resident onboarding, readiness tasks and notifications
9-10	Maintenance, contractor workflow, finance ledger and dashboard integration
11	End-to-end integration, concurrency tests, security tests and pilot data migration
12	Pilot evaluation, defect correction, performance testing and accessibility review
13	Final deployment, dissertation evidence, demonstration and presentation

Evaluation Plan

Metric	Measurement
Placement time	Median time from completed request to first suitable offer
Booking correctness	Number of confirmed double bookings under concurrent test load
Hold correctness	Percentage of holds released at the configured expiry time
Matching quality	Placement officers' relevance rating for the top three recommendations

Metric	Measurement
Room readiness	Percentage of arrivals where all required tasks finish before move-in
Maintenance response	Time from issue submission to contractor acceptance
Reconciliation accuracy	Difference between system ledger and verified test statements
Manual effort	Number of external messages or spreadsheet updates required per booking
Usability	Task-completion rate and System Usability Scale results from pilot users

The critical engineering acceptance criterion is **zero confirmed double bookings** during controlled concurrent-reservation tests.

Biggest Technical Risks

Risk	Mitigation
Booking race conditions	Use database transactions, unique constraints, idempotency keys and concurrency testing from an early sprint
Overcomplicated role permissions	Define a fixed MVP permission matrix before coding
Matching engine becomes subjective	Separate mandatory constraints from scored preferences and display explanations
Financial scope expands into accounting	Implement an operational sub-ledger and export; do not build full accounting
Six modules fail to integrate	Agree OpenAPI contracts, shared identifiers and event schemas during week 2
Pilot requests excessive customisation	Freeze the assessed MVP and place additional requests in the product backlog
Personal-data exposure	Use synthetic data by default, encrypt sensitive files and apply least-privilege access

Why the Professor Might Reject It

“It is another property-management CRUD application.”

Defence: The assessed technical contributions are transactional room holds, concurrent booking protection, an explicit booking state machine, explainable constraint-based placement matching, dependency-based room readiness and multi-party financial reconciliation. The demonstration will include simultaneous attempts to hold one room, automatic hold expiration and recovery from a failed preparation task.

“Competitors already exist.”

Defence: Existing relocation, co-living and property-operations platforms confirm market demand. The project does not claim to invent property software; it tests a narrower product for small local relocation agencies managing mixed internal and partner inventory.^{[2][5][1][7]}

“The team cannot obtain real data.”

Defence: The platform is usable with manually entered inventory and CSV migration. Development uses fictional personal records, while a pilot operator can validate workflows and provide anonymised operational measurements without supplying a large proprietary dataset.

“Six people are excessive.”

Defence: Each member owns a separate production subsystem with backend logic, data structures, APIs, interfaces and tests. The end-to-end product requires coordination among identity, booking, matching, field operations, workflow automation and finance.

Final Filter: Ten-Question Checklist

Question	Answer
Can six students build substantial code?	Yes. Six integrated technical subsystems provide clear individual ownership.
Does every student have a meaningful coding subsystem?	Yes. No member is assigned only project management, testing or documentation.
Can it be built without government or private-platform cooperation?	Yes. The core product uses agency-entered inventory and standard imports.
Can data be obtained legally?	Yes. Development uses fictional data; a pilot uses only authorised operational data.
Can synthetic data support testing?	Yes. Concurrency, matching and workflow tests can use generated records, while user validation should involve a real pilot.
Does it remain useful without AI?	Yes. Its value is booking integrity, workflow coordination and financial control.
Does it solve a real problem?	Yes. Existing relocation, co-living and property-operations markets validate the need. ^{[4][6][^7]}
Could a real company deploy it?	Yes. An agency can begin with inventory, bookings, preparation tasks and payment recording.
Does it avoid requiring licensed professional judgement?	Yes. Human operators retain responsibility for legal, property and placement decisions.
Is it technically complex enough for an MSc?	Yes. It combines multi-tenancy, concurrency control, state machines, matching, offline workflows, event processing and reconciliation.

Recommended Demonstration

1. A placement officer creates a genuine-style accommodation request.
2. The matching engine filters unavailable rooms and explains its top recommendations.
3. Two placement officers attempt to hold the same room simultaneously; only one succeeds.
4. The landlord approves the winning request and the booking becomes confirmed.
5. The outgoing resident checks out, creating inspection and cleaning tasks.
6. An inspection finds damage and blocks the room from becoming ready.
7. A contractor completes the repair and uploads evidence.

8. The coordinator verifies the work, allowing the resident to move in.
9. Finance records the deposit and first rent payment and generates the landlord statement.
10. The dashboard shows the complete audit trail, room state and operational metrics.

This demonstration presents StayOps as a functioning, deployable product rather than a collection of administrative forms.

References

1. [Coliving and Room Rental Management Software - Habitamax](#) - Specialized coliving and room rental software. Manage beds, individual contracts, shared expenses an...
2. [Coliving software, built by coliving operators](#) - Streamline your shared-living operations with ColivHQ. Built specifically for coliving, our integrat...
3. [Co-Living Property Management Software on Salesforce](#) - Unit, Room & Bed Management. Manage shared apartments, rooms, and individual beds with real-time ava...
4. [9 Global Mobility Software for 2026: Features & Benefits](#) - Looking for cutting-edge global mobility software? Explore our curated list of the 9 innovative solu...
5. [Relocation and Mobility Managers - ReloQuest](#) - Scaling global managed programs with technology, flexibility and the human element.
6. [Home - ReloQuest](#) - ReloQuest® provides the only transparent cloud-based sourcing platform to search and book temporary ...
7. [Breezeway - Property Care, Operations & Messaging Platform](#) - Breezeway's property care, operations and messaging platform helps coordinate, communicate, and veri...
8. [Planned and reactive maintenance | PMS features](#) - Property operations is where the promise made at booking becomes the stay that is actually delivered...
9. [Studentpad - Student Accommodation Software in Ireland](#) - Market leading student accommodation management software tailored to the unique needs of universitie...